

Compound Interest → Iterative Soln

The formula for compound interest with annual contributions is:

$$A = P \times \left(1 + \frac{r}{100}\right)^n + C \times \left(\frac{\left(1 + \frac{r}{100}\right)^n - 1}{\frac{r}{100}}\right)$$

where:

- **A** is the final amount after **n** years.
- **P** is the initial principal (initial investment amount).
- **r** is the annual interest rate (as a percentage).
- **C** is the annual contribution made at the end of each year.
- **n** is the number of years the money is compounded.

The formula consists of two parts:

1. $P \times \left(1 + \frac{r}{100}\right)^n$: This calculates the compound interest on the initial investment alone.
2. $C \times \left(\frac{\left(1 + \frac{r}{100}\right)^n - 1}{\frac{r}{100}}\right)$: This part accounts for the effect of the annual contributions, compounding each year along with the interest.

Soln Breakdown:

def compound_interest(principal, rate, contribution, years):

 r=rate/100

#Calculate compound interest on the initial investment

 Initial_compound = principal * ((1+r) ** years)

#Calculate compound interest on the annual contribution

 Contribution_compound = contribution * (((1+r) ** years - 1) / r)

#Sum the two parts of the total amount

 Total = initial_compound + contribution_compound

 Return round(total,2)